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ASSIGNMENT 2 DISTRIBUTION GRID SIMULATION – REPORT

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1 METHODOLOGY

1.1 DATA EXPLORATION

We first construct the data set B grid using PandaPower. And explore the dataset using a test allocation of load and generation indices that match the bus numbers. The stacked load and generation data plots reveal a typical household power demand curve with peaks in the morning and evening, and a typical PV generation curve Figure A.1. It is crucial to acknowledge that the peak generation is considerably greater than the peak load. We then calculate and plot the net imported and exported power at each household, taking into account the self-consumption of the generated power. To better understand the max line loads, we again use a stacked plot, where the top graph shows power import and power export across all time steps Figure A.2. We can now observe that both the generation and load peaks are reduced. A heat map of the largest exporters and importers of power is used to determine how to optimize load and generation allocations Figure A.3. For example, the generation on Bus 12 could be relocated to Bus 6, and the largest exporters could be spaced out evenly to reduce the load on the lines. The time series simulation of this test allocation revealed a sum maximum line loading of 124%. This was further reduced to 118% with a second manual guess, see Figure A.4. Nonetheless, the number of potential allocation combinations is $17! \times 17!$, necessitating a more intelligent method to achieve further optimization.

1.2 HEURISTIC ALGORITHMS

The Greedy Round Robin (GRR) heuristic is used to find an initial allocation for the optimization process. It is a low-cost, deterministic algorithm to quickly decrease max line loading, ranking every house based on peak net export and placing generation-and-load pairs with an even-odd interleaving to spread out high exporters.
The Smart Seed (SS) heuristic is a second deterministic, greedy algorithm. It pairs the best generation-and-load profiles and assigns the optimal pair to the weakest bus. In contrast, the least optimal pair is assigned to the strongest bus. The strength of each bus is determined using the Short-Circuit Impedance Method.

1.3 Genetic Algorithm

The Genetic Algorithm (GA) is a metaheuristic optimization technique inspired by the principles of evolutionary biology, operating on a population of potential solutions (generation and load allocations) rather than a single solution [1]. We selected the GA for this optimization task due to its ability to effectively navigate the complex search space. Furthermore, it provides robust global optimization capability [2].
Our implementation follows the structure to evolve a population of generation and load allocation pairs, illustrated in Figure A.5. The process begins by generating a population of allocations, using an initial guess from our heuristic analysis and other randomly generated guesses. Each individual is evaluated by a fitness score, which is calculated using the sum of maximum line loading per time step $Fitness = \frac{1}{Sum\ of\ max\ loading}$. A tournament selection randomly chooses two individuals to become parents, selecting the one with the highest fitness score. Using an Ordered Crossover, a random segment of first parents' allocation is copied to create a new child, with the rest being filled with the second parents indices. To prevent the algorithm from getting stuck and explore new areas, random changes in the child are introduced using Swap Mutation. The Elitism strategy ensures that the best solution is preserved by carrying over the best individual to the next generation. The remaining slots are then filled with the generated children. This cycle repeats for a predefined number of generations, progressively improving the solution.

1.4 Tuning

Firstly, the population size can be increased to search in a wider space, lowering the risk of a local optimum. By changing the tournament size, the diversity of the population is adjusted. Where a too large size would always select the best parent, creating the risk of reaching a local optimum. Furthermore, the mutation rate can be increased to add more exploration. To determine the number of generations, we decided to stop when the best solution flat-lines for around 10 generations, assuming to have found an optimum.
Due to the stochastic nature of the GA, it is difficult to find the optimal parameters. To avoid this, a JSON file with a seed number was generated for each simulation. With the seed, we could initiate the exact same environment while tweaking the parameters, giving an objective look at the resulting change.

2 Results

After having problems running the genetic algorithm for the competition, we were able to compute the entire algorithm for Grid C. The best result we achieved for the sum of maximum line loading is 90.50% using the parameters mentioned in Figure A.6. It shows that at first the improvements are big, but with an increasing number of generations the system converges and ends after 60 generations.

3 Discussion

The model combines a low-cost heuristic approach with a metaheuristic optimization algorithm that finds a local optimal solution. The solution found is highly determined by the starting population of the GA. Since both heuristic approaches generate a better starting point than random, the final result might initially seem to benefit from including both in the starting population. Surprisingly, this was not the case. In every simulation, the GA performed best when only the GRR was included. This makes sense: combining two similar heuristics decreases diversity and leads to premature convergence, yielding a worse result. The GRR and SS algorithms

most likely produce allocations with similar topology.
To counteract this, the heuristics in the starting population must be structurally different to cover more of the search space. Both heuristic algorithms treat generation and load as being tied together, making them correlated. For the GA to explore more possibilities, one of the heuristic algorithms should, for example, allow for independent placement of generation and load. This increases the computational cost of this heuristic algorithm, but likely gives the GA a better starting point, with a net decrease in total computation. Additionally, the GRR could be combined with an entirely different type of heuristic, e.g. GRASP, which is stochastic instead of deterministic at every step, and gives multiple distinct local optima. These two approaches are likely to yield an improved solution.
Tuning could also be improved. Parameters that work best for five generations are not necessarily the optimal ones for one hundred. Take, for example, the mutation rate; a low mutation rate works well with a diverse starting population, since it stabilizes convergence. If the initial population is too homogeneous and the mutation rate is low, premature convergence is very likely because there is a low probability of introducing new structural patterns. An increased mutation rate increases exploration, but with a lower number of generations, there is not enough time to exploit the new structure, effectively making the mutations into noise.

A Appendix

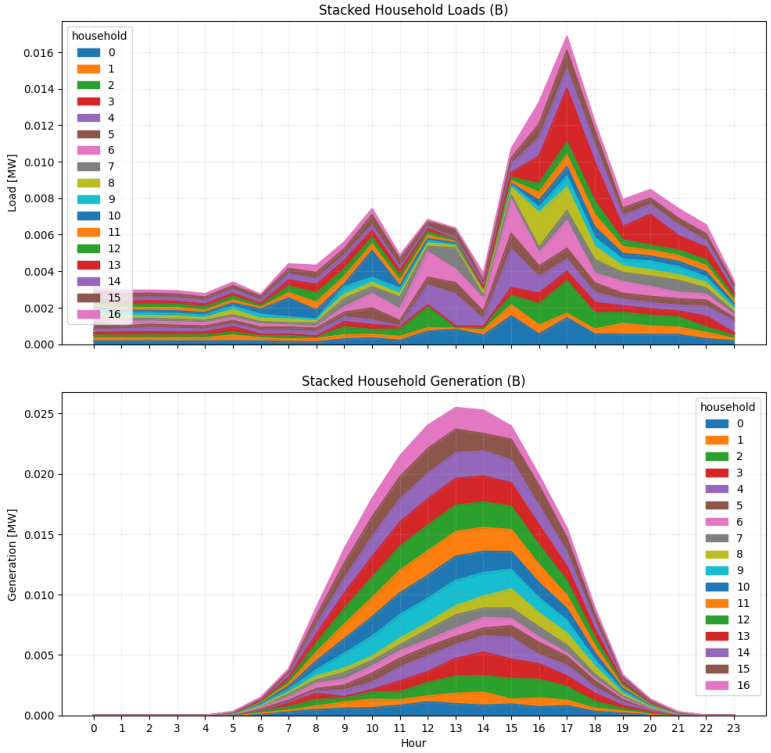


Figure A.1: Stacked Household Load and Generation Data

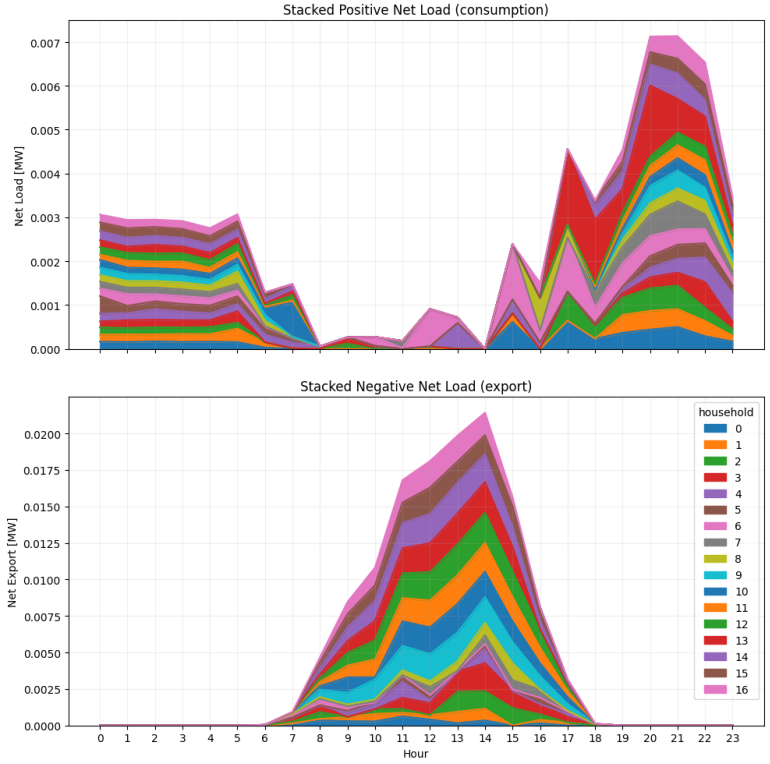


Figure A.2: Net Load per Household

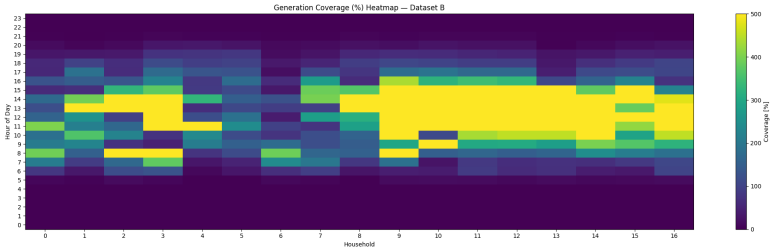


Figure A.3: Heatmap of Power Export and Import in Each Time Step

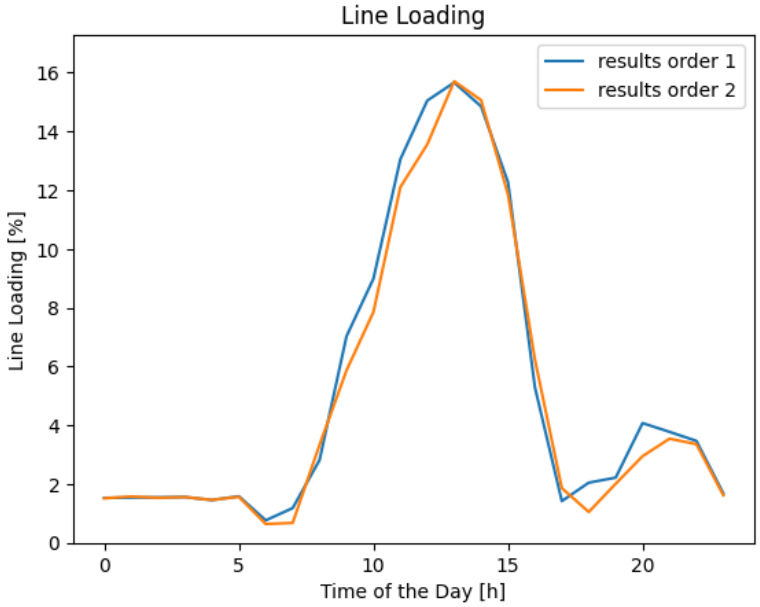


Figure A.4: Maximum Line Loading for Both Manual Guesses

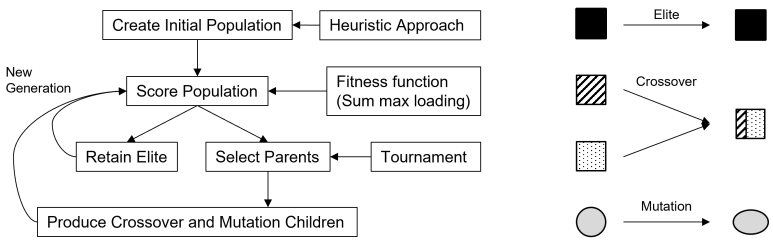


Figure A.5: Structure of Genetic Algorithm

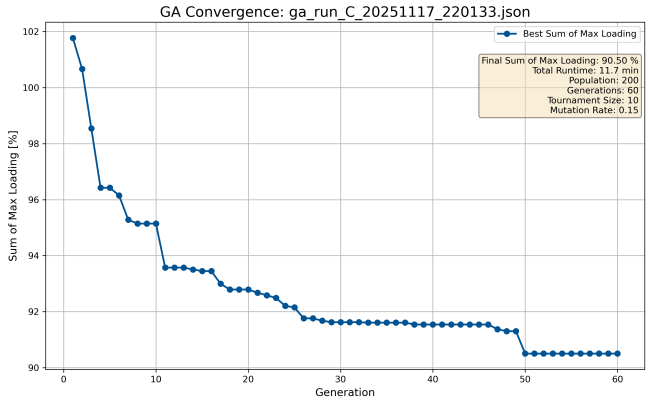


Figure A.6: Convergence Graph of Genetic Algorithm

REFERENCES

[1] Mathworks, "What is the genetic algorithm?." https://de.mathworks.com/help/gads/what-is-the-genetic-algorithm.html. Accessed: 2025-11-11.

[2] AnalytixLabs, "A complete guide to genetic algorithm - advantages, limitations & more." https://www.analytixlabs.co.in/blog/genetic-algorithm/#Advantages_of_Genetic_Algorithm. Accessed: 2025-11-11.